

Natural Gas Dehydration

The Physics and Mechanics
of TEG Absorption

Prepared by Engineer Sajjad k. Hano



Defining Success by the Thermal Buffer

Critical Concept: Thinking in Reverse — Design the plant by working backward from the required thermal end-state.



Water-Saturated
Temp



Dew Point
Depression



Dehydrated
Dew Point

40-140°F



The Two Philosophies of Water Removal

Adsorption

- Phase: Granulated Solid Bed
- Mechanism: Microscopic Pores & Capillary Openings
- Architecture: Dual-Tower Switching System
- Regeneration: High-Temperature Gas Heater



Absorption

- Phase: Liquid Solvent / Dehydrating Agent
- Mechanism: Chemical Affinity for Water
- Architecture: Counter-Current Flow Tower
- Regeneration: Continuous Reconcentration



The Survival Criteria for a Chemical Solvent

Critical Concept: Questioning Assumptions — A hydrophilic liquid is not enough. The solvent must survive immense thermodynamic stress.

- ✓ High water absorption efficiency
- ✓ High decomposition temperature
- ✓ Low vaporization losses
- ✓ Easy and economic separation and regeneration
- ✓ Non-corrosive and non-toxic to the system

The Glycol Arena and the Supremacy of TEG

EG

Superior Dew Point
Depression: 40–140°F

Regeneration Capability:
~99% Lean

High Decomposition
Temperature

Low Operating Cost &
Vaporization Loss

Operating Pressure:
25–2,500 psig

Operating Temperature:
40–160°F

DEG

Superior Dew Point
Depression: 40–140°F

Regeneration Capability:
~99% Lean

High Decomposition
Temperature

Low Operating Cost &
Vaporization Loss

Operating Pressure:
25–2,500 psig

Operating Temperature:
40–160°F

TEG

Superior Dew Point
Depression: 40–140°F

Regeneration Capability:
~99% Lean

High Decomposition
Temperature

Low Operating Cost &
Vaporization Loss

Operating Pressure:
25–2,500 psig

Operating Temperature:
40–160°F

T4EG

Superior Dew Point
Depression: 40–40°F

Regeneration Capability:
~109% Lean

High Decomposition
Temperature

Low Operating Cost &
Vaporization Loss

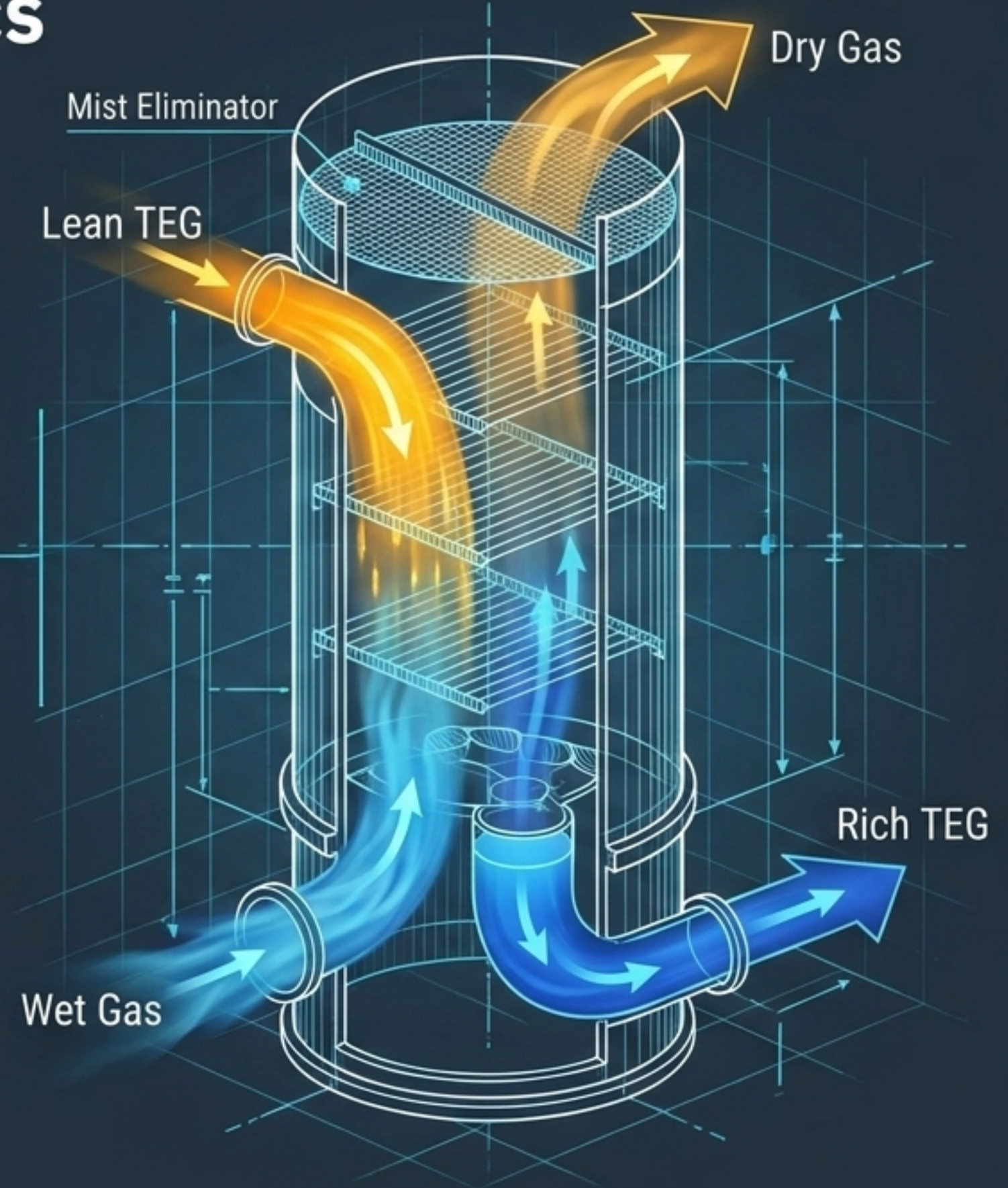
Operating Pressure:
25–2,500 psig

Operating Temperature:
40–160°F

The Counter-Current Dynamics of the Absorber

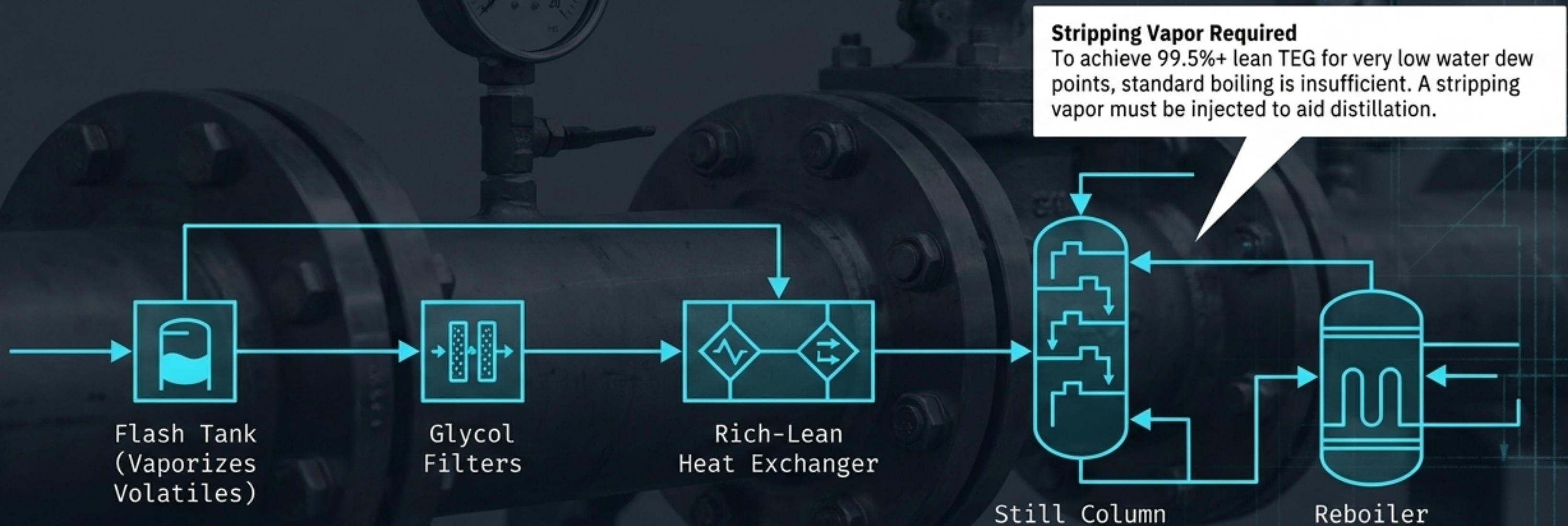
Design Criterion: Circulation Rate

Requires 2 to 6 gal TEG per lb H₂O removed for most field absorbers.



Breaking Thermodynamic Limits in Regeneration

Critical Concept: Unconventional Thinking — Standard distillation hits a thermodynamic wall. Extreme purity requires an engineered workaround.



Translating Chemistry into Physical Steel

Critical Concept: Evaluating Meta-Thinking — Reconciling theoretical equilibrium models with the empirical reality of structural packing.

$$h = (HETP) \times (N)$$

The Theoretical Model (N)



Number of theoretical stages where vapor and liquid reach perfect equilibrium

Height Equivalent to a Theoretical Plate (HETP)

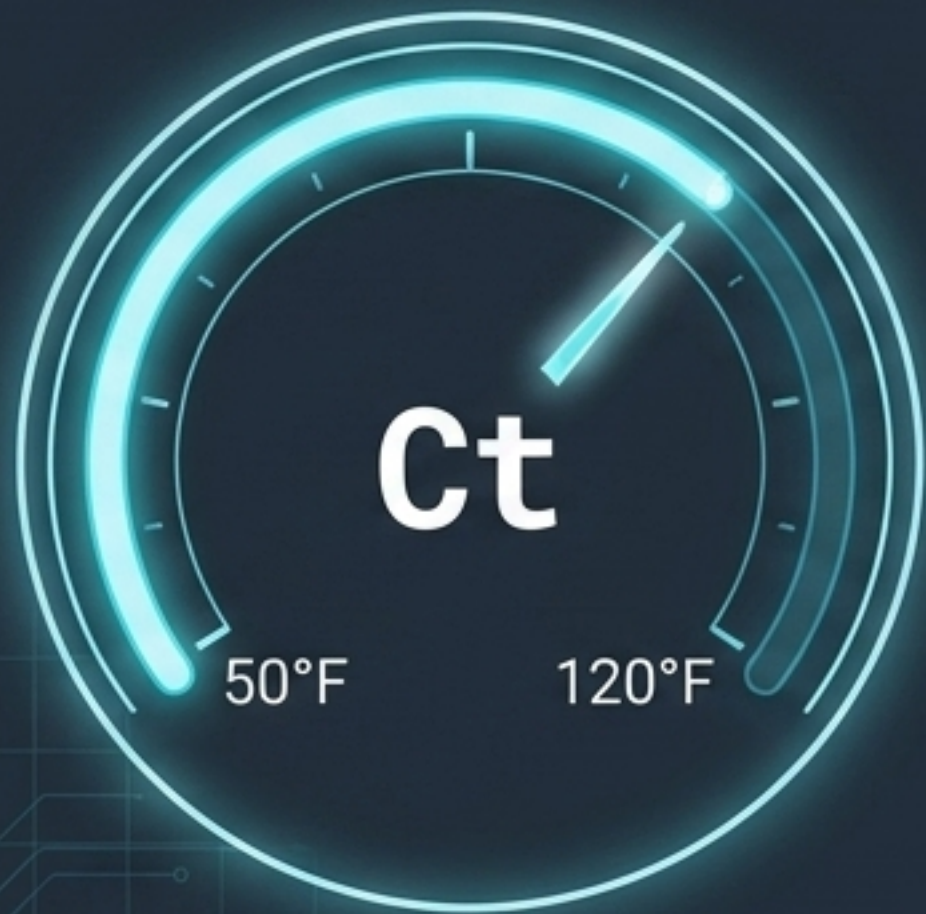


Empirically determined as 5 ft (1.5 m) per stage for both random and structured packing.

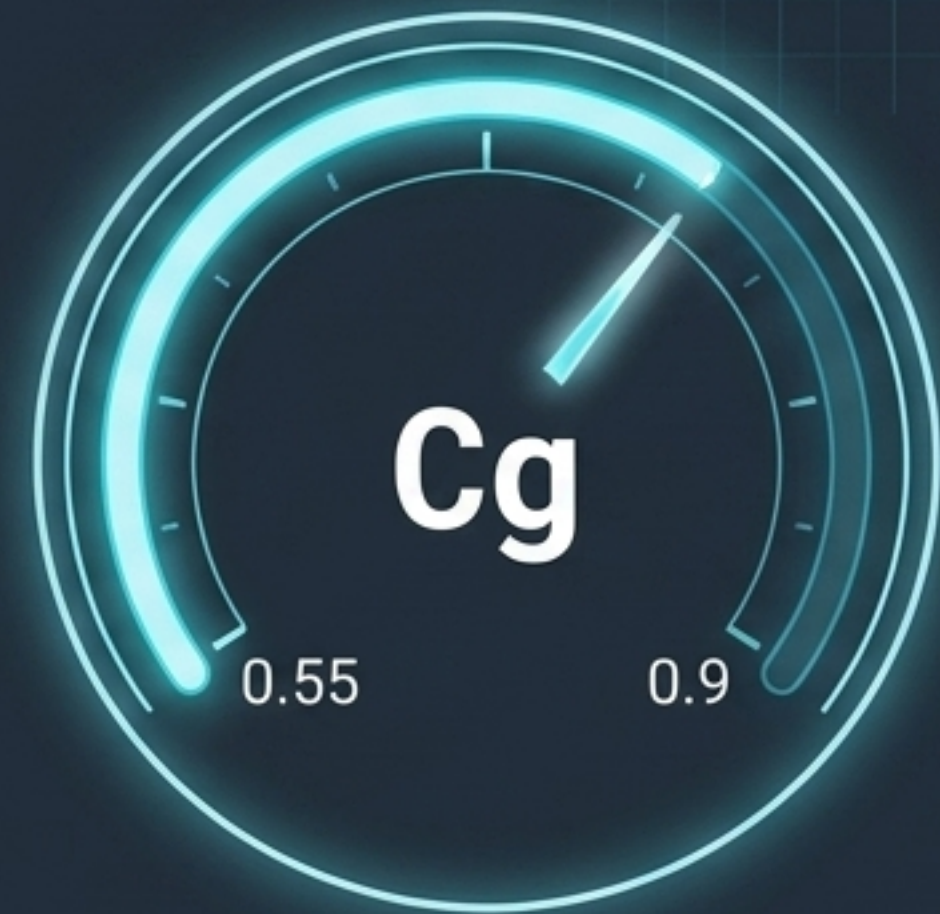
Empirically determined as 5 ft (1.5 m) per stage for both random and structured packing.

Capacity Bends to Gravity and Temperature

$$Q_o = Q_s(C_t)(C_g)$$



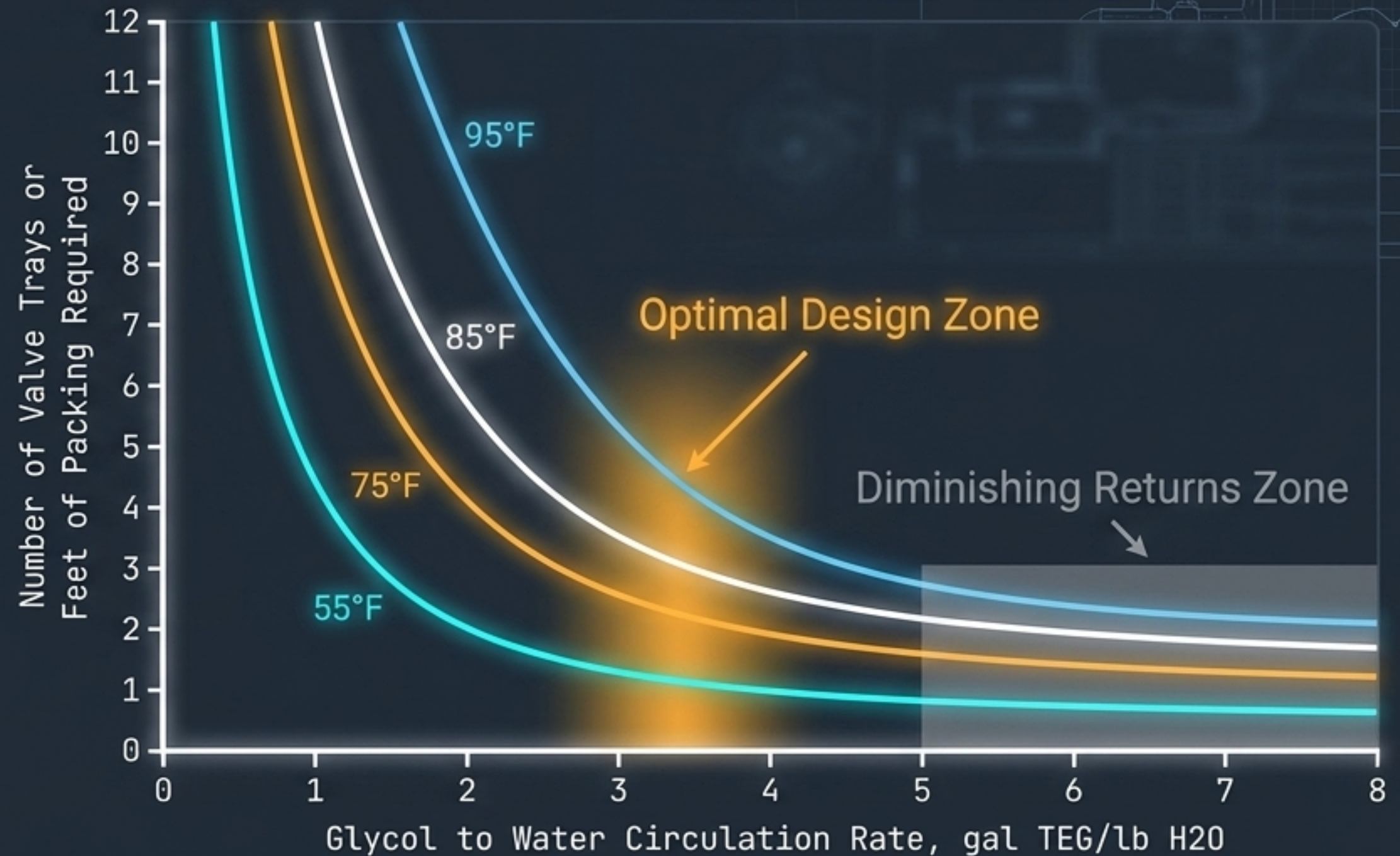
$$C_t = 0.601 T^{0.1103}$$



$$C_g = 0.6429(y_g)^2 - 1.6298(y_g) + 1.829$$

The Ultimate Engineering Trade-Off

Critical Concept: Reading Between the Lines — Beyond the optimal elbow, increasing pump output yields zero benefit. You must add physical steel.





A Perfect Harmony of Process Design

Thermodynamic goals dictate the chemical solvent, which forces the mechanical architecture, culminating in the precise dimensions of steel.